

21ECP401L-MAJOR PROJECT
SMART AGRICULTURE FOR RESOURCE
OPTIMIZATION

A PROJECT REPORT

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DECLARATION

We hereby declare that the Major Project entitled “**Smart Agriculture for Resources Optimization**” to be submitted for the Degree of Bachelor of Technology is our original work as a team and the dissertation has not formed the basis of any degree, diploma, associateship or fellowship of similar other titles. It has not been submitted to any other University or institution for the award of any degree or diploma.

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ABSTRACT

Agriculture is an important aspect in ensuring food security and the sustainability of the global population. However, plant diseases have been known to contribute to a significant reduction in agricultural productivity and economic losses for farmers. Conventional approaches to plant disease diagnosis and pesticide dosage application are manual and time-consuming. To overcome these challenges, this project proposes the use of an AI-based Drone System for Plant Disease Diagnosis and Pesticide Dosage Recommendation. Real-time identification of plant diseases and effective pesticide control are essential in contemporary agriculture. This paper presents an autonomous drone precision agriculture system using deep learning, cloud computing, and IoT technology for plant disease prediction, smart pesticide selection, optimal dosage calculation, and precise spraying. The ESP32-CAM module on the drone takes high-resolution images of leaves during autonomous flight and transfers them to Firebase Cloud Storage over Wi-Fi. The cloud-hosted Convolutional Neural Network (CNN) model analyzes the images to identify plant diseases and predict their severity levels with high accuracy. Depending on plant disease and severity percentage, the system automatically suggests the best pesticide and computes the optimal dosage based on pre-defined agronomic requirements.

Sustainable Development Goal

SDG 2: Zero Hunger – AI-based early disease detection using drones improves agricultural productivity and food security by lowering crop losses.



SDG 9 - Industry, Innovation and Infrastructure – Promotes innovation through AI and drone technologies.



SDG 12- Responsible Consumption and Production – Reduces overuse of pesticides. Optimized pesticide usage

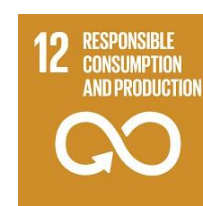


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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ESP32	Espressif 32-bit Microcontroller
GPIO	General Purpose Input Output
DC	Direct Current
AC	Alternating Current
PCB	Printed Circuit Board
ACK	Acknowledgment
IDE	Integrated Development Environment
Wi-Fi	Wireless Fidelity
IoT	Internet of Things
SDA	Serial Data Line
SCL	Serial Clock Line
VCC	Voltage Common Collector
GND	Ground
RX	Receive
TX	Transmit
PWM	Pulse Width Modulation

CHAPTER 1

INTRODUCTION

1.1 General

Agriculture is a major part of the world economy and assures food security. However, plant diseases are one of the most serious issues influencing the quality and quantity of crops. If diseases affecting plants are not identified at early stages, they can easily spread to other plants and lead to a significant reduction in the overall productivity levels and increase the use of chemical pesticides. Therefore, it is essential to identify plant diseases at an early stage to make agriculture efficient and prevent losses of money. Determining the diseases affecting plants requires farmers or agricultural experts to examine the plants physically. These processes are time-consuming, require a lot of labour, and rely solely on expert knowledge, making them ineffective for large scale agriculture. Also, manual analysis is usually incorrect due to similar symptoms of diseases, a similar environment, and human error. Recent breakthroughs in machine learning (ML) and deep learning (DL) have opened new possibilities in the field of smart agriculture. Convolutional Neural Networks (CNNs) are a powerful deep learning technique that can automatically identify the relevant features from leaf images to achieve image-based plant disease identification effectively. The algorithms correctly identify unhealthy and healthy plants in scenarios in the real world that are hard to comprehend. Besides, the use of drone technology has helped monitor crops [1-3].

Agriculture is the way that people get food. Lots of people around the world depend on agriculture. It helps countries grow and stay stable. Agriculture is very important because it gives us food and helps us survive. Agriculture has some big problems. One of the problems is crop diseases. These diseases can spread quickly across fields and hurt a lot of crops in a short time. This means that farmers do not get much food from their crops and the food is not as good. As a result, farmers lose money. There might not be enough food, for everyone. So, it is very important to keep crops healthy and stop diseases from spreading. This is necessary for agriculture to be sustainable[4].

1.2 Need for the Project

Farmers usually walk through their fields to check on their crops and use what they know from experience to find and treat sickness in the crops. They have been doing it this way for a time. This way is hard work, especially for big farms. Sometimes farmers do not see the sickness in

the crops correctly. They might find out late. Because of this they often only see the sickness when it is very bad. This hurts their crops. Costs them money. Farmers also might use chemicals to try to make the sickness go away which costs more money and is bad, for the earth and people. We need a system that helps farmers find sick plants early and make decisions. The old way isn't working. Farmers need help with their crops. Diseases in plants are a problem. We need to figure out how to spot plant diseases early.[5-6]

1.3 Objective

The main goal of this project is to make a system that uses drones and AI to find and treat crop diseases. Drones with cameras will be used by this system to take pictures of crops as they grow. A computer system that uses intelligence to find diseases and see how bad they are, even when they are just starting and can't be seen by people, will then look at these pictures. The project also wants to give farmers advice on what kind of pesticides to use and how much to use, based on what farmers do. By doing all this the system wants to help farmers have crop loss grow more food and make good decisions about their crops. The system is, about using drone technology and artificial intelligence to help with crop disease detection and management. This information helps farmers detect the presence of a disease early on and take action to prevent its spread. Combining drone images with cloud computing and AI models used for detecting diseases is a novel approach suggested for the precision farming domain. This technology not only helps in the detection of diseases, but it also helps farmers use the right amount of pesticides, is less harmful to the environment, and makes farming a sustainable process. To combat these issues, the proposed project is a system that uses AI for predicting plant diseases with help with automatic pesticide dosage and spraying from a drone

1.4 Scope of the Project

This project is important for farming today for large farms where its hard to keep an eye on crops all the time. The system helps farmers check their crops health regularly find diseases and use the amount of pesticides. This makes farming easier and more efficient. It also helps reduce chemical use saving farmers money and benefiting the environment. Farmers can get needed information on their phones. Take quick action. In the future this project can do more like analyzing real-time data predicting weather and automatically spraying pesticides. This will make farming more modern and sustainable. The project will improve farming making it more efficient which is great for farmers and the environment. Farmers and the environment will both benefit from this project. The entire system is connected to the Sustainable Development Goals (SDGs) of zero Hunger (2-SDG) and

responsible consumption and production (SDG 12). It reduces the consumption of pesticides and supports innovative ideas through the application of AI and drone technology (SDG 9). It is possible to always monitor crops' health by making the process large-scale and obtaining aerial images in real time with the help of high-resolution camera drones. The approach relies on a CNN model, which has been trained using the PlantVillage dataset, to perform automatic identification of plant diseases based on images of plant leaves. Technology uses a well-organized agricultural dataset to make recommendations for the most suitable pesticide and dosage to use based on the disease identified, as well as the severity level determined directly from the image. Also, a drone spraying system calculates the amount of water and insecticide needed for the work based on the size of the field. The system combines drone-based image collection, cloud storage, deep learning for disease detection, pesticide suggestions, and automated spraying.

CHAPTER 2

LITERATURE REVIEW

2.1 General

Plant disease detection is a deal in farming because it affects how much food we can grow and how safe that food is to eat. In the past they had to look at the plants by hand and use ways to look at pictures but this was not very accurate and took a lot of time. Now build systems that work better on their own because we have things like intelligence. A lot of studies have shown that using Machine Learning and Deep Learning together can help us find diseases more accurately [7][8]. This is very important for farming to go well. This is making it easier to find plant diseases. These days, a lot of people use machine learning and deep learning to find plant diseases. A lot of scientists have made systems that can look at pictures and tell you what disease a plant has. This helps improve farming.

2.2 Deep Learning Approaches (CNN Models)

Finding plant diseases is becoming thanks to learning especially with Convolutional Neural Networks. This is a way to find plant diseases. Researchers like Sujatha and others have found that using machine learning and deep learning together makes it easier to find things. They do this by removing features and making decisions that are structured. For example Zainab and Mahum also learned that Convolutional Neural Networks work best when they are trained on datasets that have been cleaned and prepared correctly. This includes resizing all the pictures and making them all the same size. People often use models like AlexNet, VGG and Resnet to figure out what diseases plants have. These models are great at taking pictures and finding things in them. This means that it is faster and more accurate because people do not need to do that work to find the diseases. Convolutional Neural Networks is very useful in plant disease detection. Then came deep learning. It was much better at diagnosing what disease a plant has. This is where Convolutional Neural Networks shine. According to some studies like the ones [9-11] with a data set like Plant Village they can get it right over 95 percent of the time. Deep learning models can look at pictures. Find out what is important for themselves so it is not necessary for people to do it manually and this makes it easier to use in complicated places. Some people have also built systems that use clouds to look at pictures of plants and tell what disease they have. This means the pictures can be sent to a computer in the cloud and the answer can be sent back. This is good because you can handle a lot of pictures at the time and all the

information can be stored in one place. Most of these systems only tell what disease the plant has. They do not provide any advice on what to do next such as what kind of pesticide to use and how much. Convolutional Neural Networks are really good, at finding plant diseases.

2.3 Drone and UAV-based Monitoring Systems

Agriculture is getting better with technology. People are now using drones or Unmanned Aerial Vehicles for watching over farms. These drones have cameras that can take pictures of crops quickly over large areas. The pictures are then looked at with the help of computer programs to find diseases in the crops. Agriculture is also using cloud computing and internet technologies to deal with a lot of data at the time. Some studies, like the ones done . show that it is important to have a system in place to handle problems like slow connections and not being able to process information fast enough. All these new technologies are helping to create systems that can watch over farms in time and can be used for big areas of land. This is good, for agriculture and it is using drones and other technologies to make things better. People have also used drones to look at plants and figure out what disease they have. Drones can take pictures quickly and can get very close to the plants. Some people have even used drones to spray pesticides. This is still not very common. What is missing is a system that can do everything: Let the drone move around on its own

- Look at pictures and figure out what disease the plant has
- Say how bad the disease is • Say what kind of pesticide to use and how much
- Control the spraying of pesticide Let the farmer see what is going on from a distance using a website The new system is going to fix these problems by using machine learning to look at pictures of plants, a cloud-based system to make decisions and a drone that can spray pesticides on its own. Farmers can see what the system is doing on a website that tracks what diseases are present in the plants and when the drone is spraying in real time. “This is progress for farming, as it combines artificial intelligence, internet-connected devices and drones into a single user-friendly system that works well. The system is also good because it's efficient and can be used by a lot of people. The new system uses machine learning to analyze plant images. It has a cloud based decision making system and a drone platform that can spray pesticides[12-15].

2.4 Challenges

- Plant disease detection has improved, but many challenges still remain.
- Most AI models are trained using controlled datasets such as PlantVillage.
- These models may not perform well in real-world farming conditions due to differences in lighting, background, and environment.
- Many existing studies focus only on disease identification.
- Researchers often do not provide recommendations about:
 - Suitable pesticides
 - Required pesticide quantity or dosage
 - Proper treatment methods
- There are additional challenges related to:
 - Cost of implementation
 - Accessibility of technology
 - Adoption by small-scale farmers
 - Deployment and maintenance in rural areas
- Future research should focus on developing models using real-world agricultural data.
- Edge computing can be used to process data efficiently and reduce response time.
- Integrating AI, IoT, drones, and automated spraying systems can improve farming efficiency.
- Intelligent systems can support both automatic pesticide spraying and farmer decision-making.
- This project aims to address these limitations by:
 - Detecting plant diseases using AI
 - Providing pesticide recommendations
 - Suggesting appropriate dosage levels
 - Using drones for smart agricultural operations

CHAPTER 3

SYSTEM ANALYSIS

3.1 Existing System

Plant disease detection is something that farmers usually do by looking at the plants and using their experience to figure out what is wrong with the plants and how to treat the problem with the plants. This way of doing things is time consuming and it requires a lot of work. It is also not very accurate especially when it comes to fields where it is hard to keep an eye on all the plants all the time. Farmers might miss the signs of plant disease. That can lead to bad things happening to the crops and a lot of money being lost because of the plant disease. Farmers might also use many pesticides because they do not know what is really wrong with the plants. This makes things more expensive for the farmers. It hurts the environment because of the plant disease detection method. There are some tools that can help farmers with plant disease detection but not many farmers can use these tools. These tools also do not always give farmers the information they need about the plant disease. They do not always tell them what to do to treat the plant disease. This makes the whole process of plant disease detection not very good and not very trustworthy, for the farmers. Plant disease detection is a deal and it needs to be done in a better way to help the farmers with plant disease detection.

3.2 Proposed System

The proposed system is a way to find plant diseases using drones and computer technology. It uses a drone with a camera to take pictures of crops in time over big farms. These pictures are sent to a computer system. The computer system uses a kind of math model to look at the pictures. This model helps find plant diseases and tells how bad they are. The system then gives advice on what kind of pesticide to use and how much. It follows farming guidelines. The results are sent to farmers through a phone or computer app. This helps farmers make decisions. The plant disease system makes it easier for farmers gives results uses less pesticide and helps farms be more efficient and greener. The new plant disease system puts together a few things to help with farming. It uses drones that can fly around on their own a computer program that can tell when plants are sick a way to pick the pesticide a calculator to figure out how much to use, a machine that can spray pesticide and a website that shows what is happening in real time. The plant disease system is talking about work in a way. It starts with using a drone with

a camera to take pictures of leaves. These pictures are then made to look at. Any bad stuff is removed. After that the pictures are looked at closely by a computer program that uses learning to find any diseases on the plant disease systems plants. When the disease is found the computer looks at how bad it's by checking how much of the leaf is affected. The plant disease system then tells us what kind of pesticide we need to use and how much of it we need to use. The drone then sprays the pesticide where it is needed. That shows everything the plant disease system is doing on a website, including what disease was found and how bad it's what pesticide to use. The drone flies over the fields on its own by following a path. This way it can see the field. Make sure it does not miss any spots. The drone has a camera that takes pictures of the field at regular times. The pictures are taken when the drone is not moving so they are very clear and good for finding diseases on the plants. The drone is always at a height, above the field, which helps it take pictures of the plants. The plant disease system uses the drone and the camera to monitor the fields and find any diseases. It does this by using the drone to take pictures of the plants and then looking at those pictures to find any problems. The plant disease system is a tool for farmers to find and treat plant diseases quickly and efficiently.

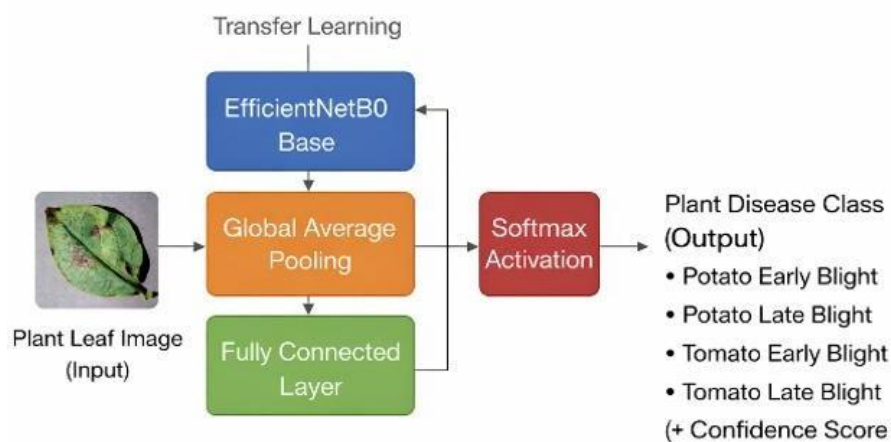


Figure 3.1. Convolutional Neural Network Architecture

Each picture has information like where it was taken and when. This helps to make a map of the field. Shows where the sick plants are. The pictures are made smaller and compressed before they are sent. This helps to save bandwidth and makes the upload faster. The pictures are then sent to Firebase Cloud Storage using Wi-Fi. They are then looked at using special computer programs to check for disease. If the internet is not working the pictures are stored on the drone. They are sent again when the internet is working. The drone also has a system to

watch the battery level and go home if it needs to. This makes the drone safe to use. Also helps in monitoring the plants. The information help, in identifying the sick plants.

3.3 Advantages of Proposed System

The proposed system has benefits compared to ways of finding plant diseases. It helps find crop diseases early and accurately using computer models. This early detection helps prevent damage to crops and increases the harvest. Drones can quickly check farms which reduces the need for workers and saves time. The system also gives advice on what type of pesticide to use and how much which helps reduce much chemical use. This saves money, protects nature. Farmers can make choices with real-time analysis and easy access through phone or computer apps. Overall the system helps farming be more precise get more done and be good for the earth. It supports farming in a way thats good for the planet. The drone is very useful for farms. It can look at the field by itself. Help find sick plants quickly. This is the start of using computers to help farms work better. The autonomous field scanning and image acquisition module helps farmers. It enables monitoring of large agricultural areas using a GPS-guided drone platform. The drone platform and the autonomous field scanning and image module work together to make farming easier. The cloud-based plant disease prediction module looks at pictures that the drone takes and sends to Firebase Cloud Storage. When a picture is saved in the cloud a special computer program called a Convolutional Neural Network model gets the picture. The Convolutional Neural Network model looks for things like color texture. What the sick parts of the plant look like. This helps figure out what disease the plant has. The plant disease prediction module tries to figure out how bad the disease is. It does this by looking at how much of the leaf's infected. The result includes the name of the disease the plant has. It also includes how sure the plant disease prediction module is and how bad the disease is. These are all saved in the cloud database. This helps recommend the pesticide and the drone can spray the right amount. This way of doing things using the cloud helps the plant disease prediction module. It can work with a lot of pictures look at them fast and be used from anywhere. This is good, for farming and growing plants. The Convolutional Neural Network used for plant disease classification has a design. This design is shown in Figure 2. The Convolutional Neural Network uses EfficientNetB0 as the network to find features. After that it uses pooling and fully connected layers to classify plant diseases. The Convolutional Neural Network is part of this process. The Convolutional Neural Network model is used to classify plant diseases. Plant

diseases are detected using the Convolutional Neural Network model. The system uses the Convolutional Neural Network model to predict plant diseases.

3.4 Limitations of the system

The proposed system has some limitations. The disease detection is only as good as the data the proposed system is trained on. If the data the proposed system is trained on is not quality then the proposed system will not work well in real life. The proposed system needs an internet connection to work properly which can be a problem in areas. The cost of the drones and the other equipment for the proposed system is high. This is not something that small farmers can easily afford. Sometimes the weather and the lighting can also affect how well the proposed system works. The cameras on the drones for the proposed system have to be of quality or the pictures will not be clear. The drones for the proposed system need to be checked and fixed regularly. You need to know how to use the proposed system which can be hard for some people. The proposed system has these limitations. We must think about the drones for the proposed system and the proposed system when we consider these things, about the proposed system.

3.5. Model Architecture

The proposed model for detecting plant disease uses a convolutional neural network based on EfficientNetB0. The main goal of this design is to have models with classification performance that can be used in real-time settings like farms. The model takes an RGB leaf image resized to $224 \times 224 \times 3$ as input. The image then goes through a data augmentation pipeline. This pipeline uses techniques including Horizontal flipping, Small-angle rotation ($\pm 8\%$), Zoom ($\pm 12\%$), Contrast-adjustment ($\pm 10\%$). These techniques help build solutions that can accommodate variations found outside during leaf capture such as changes caused by drone tilt, lighting or differences in scale. The leaf image pixel values are then normalized. This normalization procedure was used during EfficientNet pre-training. It helps the models learned weights resemble ImageNet-trained weights. This increases the likelihood of achieving classification in Figure 3.2. The primary portion of this model that creates features consists of the EfficientNetB0 architecture. This architecture considers improvements in depth, width and resolution. It uses methods such as Stage Inverted Bottleneck Convolutions (MBConv) blocks Squeeze and Excitation (SE) attention operations. These methods assist in recalibrating features on a channel basis. The result is a hierarchy of located features (disease symptoms)

created from differences, in color, texture and shape of the plant leaf. The plant leaf disease symptoms are detected using EfficientNetB0. The EfficientNetB0 model helps in detecting plant disease.

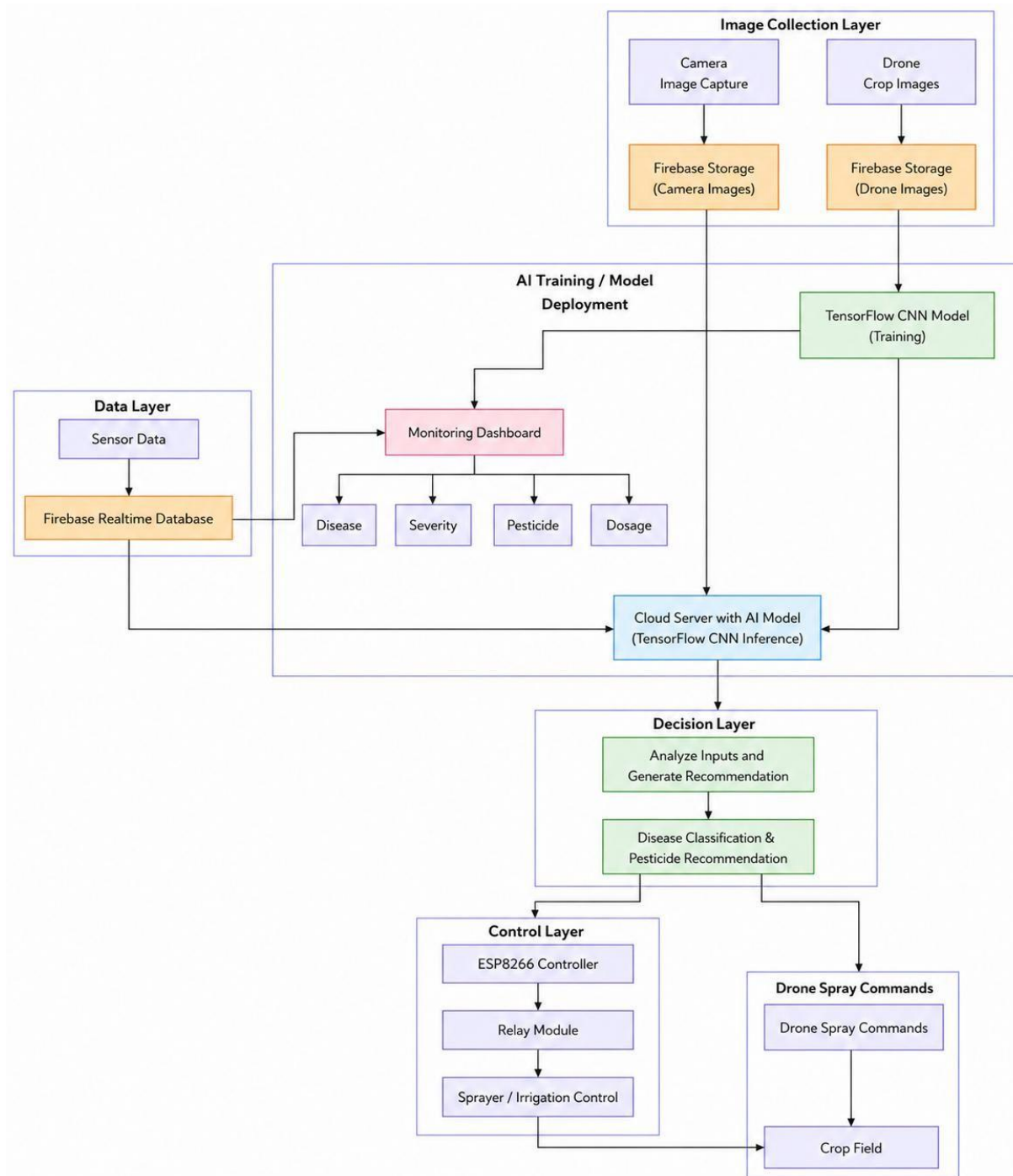


Figure 3.2. Architecture Diagram

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The system architecture of the proposed solution works like a team. It uses drones, intelligence and cloud computing to get the job done. Drones fly over fields. Take pictures of crops. This way we can check areas quickly without having to walk. These pictures are sent to a computer or cloud server using a communication network. Then the pictures are cleaned up to make them better and easier to understand. This includes making them smaller fixing the colors and removing things we do not need to see. This helps the system work better and makes sure it gets the information. After the pictures are cleaned up they are given to a computer program a Convolutional Neural Network that looks for plant diseases. This program checks the pictures for patterns, textures and colors to figure out what disease it is and how bad it is. This way we can find diseases faster and more accurately than if we were doing it by hand. When the disease is found the system sends the results to a part that gives recommendations. This part tells the farmer what pesticide to use and how much to use based on what the disease's how bad it is. This helps farmers use the chemicals in the amount so they do not hurt the environment. Finally the results are shown to the farmer on a phone or computer. The system is easy to use. It tells the farmer what disease was found what pesticide to use and how to use it. This way farmers can do something about it away. The system architecture makes sure everything works together smoothly from taking the pictures to making a decision. It helps farmers work in a way that's good for the environment and gets good results. The design methodology of the proposed system is centered around the integration of autonomous drone navigation, cloud-based learning, intelligent pesticide recommendation, automated spray control and real-time web monitoring into a single precision agriculture system. The system is developed in a fashion to ensure scalability, robustness and efficient real-time processing. First the agricultural drone autonomously scans the field using GPS waypoint navigation. An ESP32-CAM module on the drone takes high-resolution images of the leaves of the crops at fixed time intervals. Each image is annotated with metadata such as GPS coordinates and timestamp. Then transmitted to Firebase Cloud Storage over Wi-Fi for cloud-based processing. Resizing and compression are performed on the images before transmission to ensure processing and minimize latency. Once uploaded a cloud-based Convolutional Neural Network model retrieves the image. Classifies the plant disease. The model analyzes features such as color, lesions and texture irregularities

to classify the disease accurately. The system uses drones, intelligence and cloud computing to get the job done. It helps farmers work in a way that is good for the environment and gets good results. The system architecture of the proposed solution works like a team. It uses drones, intelligence and cloud computing to get the job done. The system is easy to use. It tells the farmer what disease was found what pesticide to use and how to use it. The system architecture makes sure everything works together smoothly from taking the pictures to making a decision. It helps farmers work in a way that's good, for the environment and gets good results and it uses drones, intelligence and cloud computing to get the job done. Furthermore, the severity of the disease is estimated by calculating the infected area of the leaves. Based on the classified disease and the severity percentage, the system accesses a structured pesticide database to recommend the pesticide and calculates the optimal dosage based on severity-adjusted formulas. The computed spray parameters, such as pesticide amount and spray time, are then updated in the Firebase Realtime Database. A relay-controlled pump mechanism is triggered by an ESP8266 module mounted on the drone, which continuously checks the database and sprays pesticides only over the infected area for a computed time duration. Classification of Plant Diseases Using CNN To classify plant disease within the images of the plants' use, I utilize the EfficientNetB0 model which is an accurate convolutional neural network architecture for a very large number of parameters. The EfficientNet-B0 model uses Compound Scaling to increase the Depth, Width and Size of the model in an efficient manner when extracting features while also holding down the cost of computational resource.

The CNN architecture used will consist of:

An Input Layer (224 x 224 x 3)

A Data Augmentation layer An EfficientNetB0 Backbone (Pre-trained on ImageNet)

A Global Average Pool (Pooling Layer)

A Dropout Layer (Dropout Rate = 0.3)

A Fully Connected Dense Layer with SoftMax Activation Function

Mathematically Represented Let the Input Image be:

$$X \in \mathbb{R}^{224 \times 224 \times 3}$$

$$X \in \mathbb{R}^{224 \times 224 \times 3}$$

Feature Extraction is defined mathematically as:

$$F = \text{CNN}(X) \quad F = \text{CNN}(X)$$

Classification of the result is defined mathematically as:

$$y^{\wedge} = \text{Softmax}(WF + b)$$

$$y^{\wedge} = \text{SoftMax}(WF + b)$$

Where: $y^{\wedge}$ is the Probability Vector of the Class that was Predicted W and b are Learnable Parameters. The Prediction Class of the Disease is Defined as:

$$yy = \text{argmax}(y^{\wedge})$$

$$y = \text{argmax}(y^{\wedge})$$

C. Severity of The Disease After the image of the plant has been classified this information is then used to estimate the severity of the plant disease using Color Segmentation while utilizing the HSV color space.

1. Creation of the Leaf Mask

The green pixels within the combined image will be extracted to create an area within the overall image representing the Leaf area.

$$\text{LeafMask} = \text{HSVgreen_range}(X)$$

$$\text{LeafMask} = \text{HSV green range}(X)$$

2. Lesion Identification The areas of brown/yellow within the image will be extracted which will then represent the Lesions. Concurrently, all the results, such as disease name, confidence level, severity percentage, suggested pesticide, amount, spray time, spray status, and GPS coordinates, are shown on a web interface available to farmers. This approach ensures autonomous functioning, real-time decision-making, and sustainable smart farming practices.

Data Preparation:

Preparing the data is an important step when you are trying to make a model that can predict plant diseases. The performance of the Convolutional Neural Network model really depends on how good the training data. So you must make sure the data is prepared carefully.

A lot of plant leaf pictures were collected from datasets and from pictures taken by drones in farms. The dataset has leaf samples with diseases in different lighting, backgrounds and angles. This variety helps the Convolutional Neural Network model learn features and recognize plant diseases correctly when conditions change. Each picture was labeled with the disease it had so the model can learn from it. The images were then prepared for the model. All images were made 224×224 pixels so they fit the models needs. The brightness and contrast of the images were made consistent. Noise was removed from the images to help the model learn better.

To make the model better and prevent it from being too specific more images were created from the existing ones. These new images were made by rotating, flipping, scaling moving and adjusting the brightness of the plant disease images. This helped the Convolutional Neural Network model learn patterns that're useful and work in different conditions.

The dataset was then split into three parts:

1. Training
2. Validation
3. Testing datasets.

70% Of the plant disease images were used to train the model 15% to check if the model is working well and 15% to test the model's performance. The training dataset helped the Convolutional Neural Network model learn about plant diseases. The validation dataset checked if the model was learning correctly. The testing dataset evaluated how well the trained model worked. By preparing data like this a reliable Convolutional Neural Network model, for detecting plant diseases can be made. This plant disease model can be used with drones to monitor farms and help agriculture applications.

Data Segregation:

Separating the data is one step in making a plant disease prediction model that works. When you split the dataset correctly the Convolutional Neural Network learns patterns as it is training. Can generate accurate predictions on new data. The dataset is split into three parts: training, validation and testing datasets. Usually, 70% of the images are used for training, 15% for validation and 15% for testing. The training dataset helps adjust the Convolutional Neural Network (CNN) models settings through a process called backpropagation. The validation dataset checks how well the CNN model is doing during training and helps change settings if

needed. This also helps avoid overfitting. The validation dataset is used to monitor the performance of the CNN model during training and to adjust hyperparameters when necessary. The testing dataset is not used during training. Instead, it evaluates the performance of the trained Convolutional Neural Network model on new data. A sampling method is used to ensure each dataset part has examples from all plant disease classes. This keeps the classes balanced. Makes the models predictions more stable and reliable also keep the datasets separate to prevent data leakage. No image is allowed in both training and testing datasets even if it has been changed through preprocessing or augmentation in Figure 4.1. This ensures an evaluation of the model's performance, on the Convolutional Neural Network model. The testing dataset is used to evaluate the performance of the trained Convolutional Neural Network model.

This method of separating the data makes the plant disease prediction model more dependable stops overfitting and makes sure that the autonomous drone-based plant disease detection system functions in the real world, with the Convolutional Neural Network.

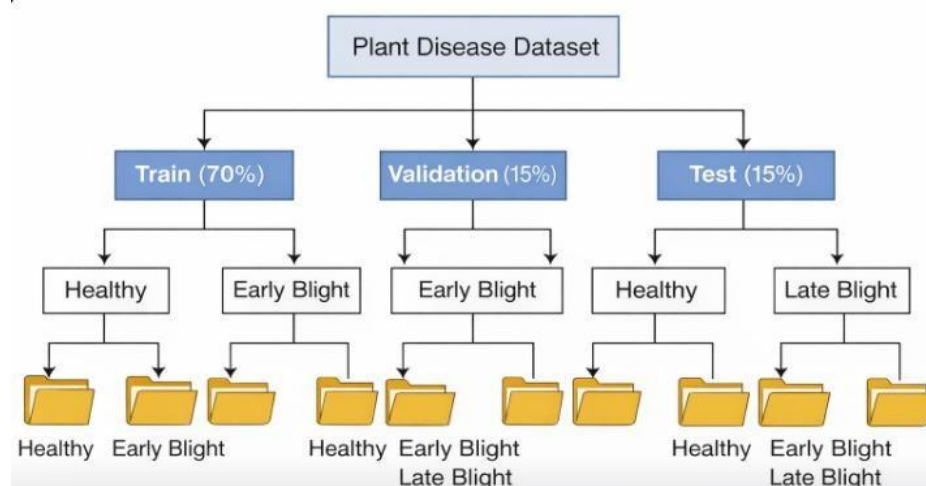


Figure 4.1. Data Segregation

The proposed system uses a dataset distribution strategy shown in Figure 3. This strategy divides the dataset into three parts:

- training
- validation
- testing This division helps to ensure that the model is trained and evaluated fairly.

The model is trained on the training subset. It is then validated on the validation subset. Finally, it is tested on the testing subset. The dataset distribution strategy is important, for model training and evaluation.

Data Augmentation:

The variability of datasets is critical to the robustness and ability of a deep-learning based model for detecting and classifying diseases of plants. The images taken from agricultural fields vary greatly with respect to light, orientation of leaves, background clutter, distance of camera, motion blur, and noise due to the environment; therefore, to improve the robustness of the model for these real-world occurrences, a comprehensive augmentation and pre-processing pipeline was executed.

(b) Pre-processing of Images

All images have been resized to a standard spatial resolution of 224×224 -pixel size that aligns with the input image size of the EfficientNetB0 architecture. Normalization of pixels and images was performed using the Efficient Net preprocessing function which scales the pixel values appropriately to be comparable to the distribution of the pretrained ImageNet weight values.

Let an input image be in a mathematical sense as follows.

$$X \in \mathbb{R}^{H \times W \times 3}$$

where X is the image and H and W are the height and width of the input image. Thus, after the resizing and normalizing process:

$$X_{processed} = f_{preprocess}(resize(X))$$

This ensures that the input images will have the same dimensionality and distribution in the training dataset.

(c) Data Augmentation Strategy

Stochastic augmentation was applied dynamically in the training process to build a diverse training dataset (the use of stochastic augmentation will reduce overfitting). The transformations used are represent

4.2 Block Diagram

The block diagram of the proposed system shows how the plant disease detection and pesticide recommendation work. It starts with a drone that has a camera. This camera takes pictures of crops in time. These pictures are sent to the system so it can do something with them.

The system does some things to the pictures first. It makes them smaller and gets rid of some noise. Then a thing called a Convolutional Neural Network model looks at the pictures to figure out if the plants are sick and what is wrong with them in Figure 4.2.

When the system knows what is wrong with the plants it tells the farmer what kind of pesticide to use and how much to use. The system shows the farmer all this information so the farmer can make decisions, about the crops. The plant disease detection and pesticide recommendation system helps the farmer take care of the crops.

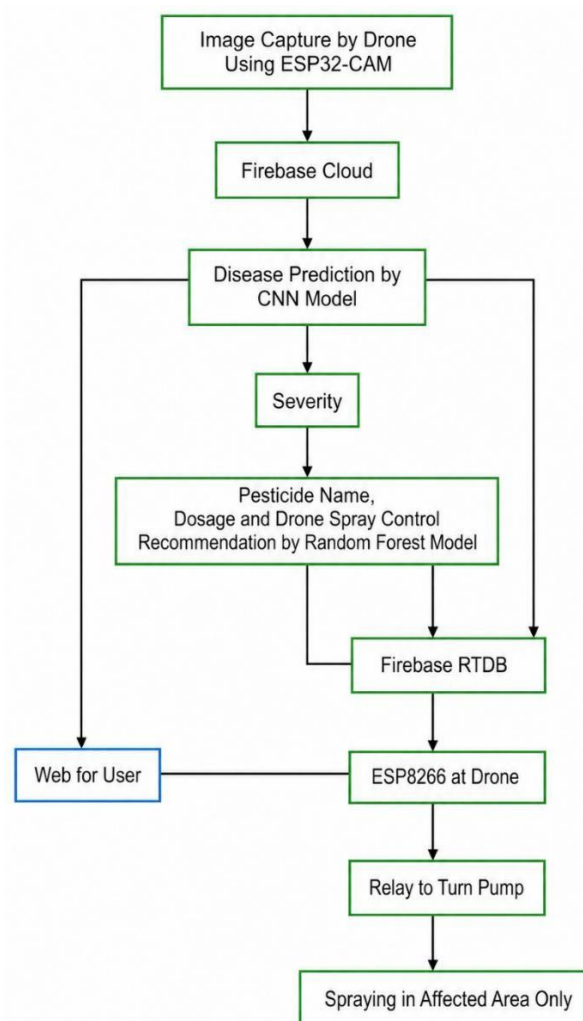


Figure 4.2. Block Diagram of the Project

4.3 Components Description

1) ESP32-CAM

The ESP32-CAM is a compact microcontroller module with an integrated camera and Wi-Fi capability. It is widely used for image capture, video streaming, and IoT-based vision applications.



Figure 4.3. Esp Cam Module

2) NodeMCU ESP8266

The NodeMCU ESP8266 is a Wi-Fi-enabled microcontroller board based on the ESP8266 chip. It acts as the main controller of the system.

It connects to the internet and communicates with the cloud database to receive control commands and operate actuators such as relays and spray motors.



Figure 4.4. Esp 8266

3) Relay Module (Single/4 Channel)

The Relay Module is used to control high-voltage devices using low-voltage signals from the ESP8266.

It acts as a switch to turn ON/OFF devices like pumps, motors, or valves.



Figure 4.5. Relay Module

4) Water Pump / Spray Motor

The DC Water Pump is used to spray pesticide or water onto crops. It is controlled using PWM signals from the ESP8266 to adjust spray intensity based on disease severity.



Figure 4.6. Water Pump

5) Firebase Realtime Database

The Firebase Realtime Database is a cloud-based database used for storing and synchronizing data in real time. It enables communication between the AI system and the ESP8266.

6) AI-Based Image Processing System

The system uses a trained deep learning model for plant disease detection. It analyzes leaf images and predicts disease type and severity.

4.4 Working Methodology

The proposed system is really simple. It does things one step at a time to figure out if plants are sick. The system tells the farmer what to do to help the plants. First a drone with a camera takes pictures of the crops. These pictures are sent to a computer so people can look at them closely. This way people can get a lot of information about the crops. When the pictures get to the computer they are made to look better. They are made smaller. Any extra things that are not needed are removed. This helps the computer understand the pictures of the crops. After that the pictures are shown to a computer program.

This program is called a Convolutional Neural Network.

The Convolutional Neural Network looks closely at the pictures to see if it can find any plant diseases. It checks the pictures of the leaves for things, like patterns and textures and colors. The Convolutional Neural Network does this to figure out what kind of disease the plant has and how bad the disease is. The system can find the disease faster and more accurately than a farmer could. The proposed system uses the Convolutional Neural Network to help the farmer take care of the plants.

When the computer finds out what is wrong with the plant it sends that information to another part of the system in Figure 4.7. This part tells the farmer what kind of medicine to use on the plant and how much to use. It does this by following the rules for farming. This helps make sure that not much medicine is used and it helps the plant get better. Finally the farmer can see the results on their phone or computer. The system tells them what is wrong, with the plant, what medicine to use and how to use it. This helps the farmer take care of the plant. Makes sure they have healthy crops. The whole system helps farmers take care of their plants and it helps them grow more food.

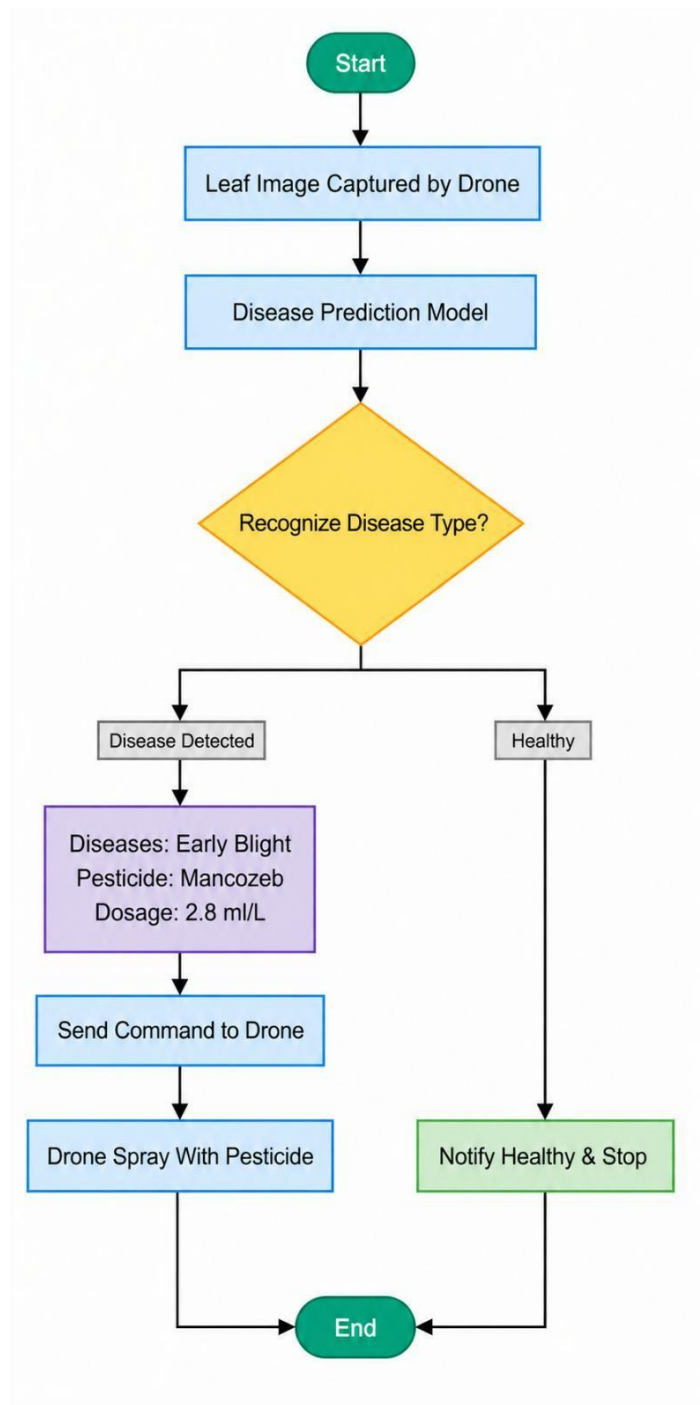


Figure 4.7. Flow Chart

4.5 Algorithm Steps

The algorithm of the proposed system starts with image acquisition. A drone with a high-resolution camera takes pictures of crops in the field. These pictures are then uploaded to the system or cloud server for processing. In the preprocessing stage the images are made smaller cleaned up and improved. This helps to make the pictures clearer. The improved images are then sent to a kind of computer program called a Convolutional Neural Network or CNN. The

CNN automatically finds details in the pictures, such as color, texture and patterns. Based on these details the program decides if the plant is healthy or not. If not it identifies the type of disease.

1. Start system initialization – Power on drone, ESP32-CAM, sensors, and establish Wi-Fi/cloud connection (Firebase).
2. Image acquisition – Capture crop images using ESP32-CAM mounted on the drone.
3. Data transmission – Upload captured images to the cloud server for processing.
4. Preprocessing – Resize images, normalize, and enhance quality for better analysis.
5. Disease detection – Apply CNN model (EfficientNetB0) to classify plant disease.
6. Severity estimation – Use image segmentation (HSV-based) to calculate infected area percentage.
7. Feature extraction – Collect inputs like disease type, severity level, and environmental data.
8. Pesticide recommendation – Use Random Forest model to predict suitable pesticide and dosage.
9. Decision making – Generate control signals based on zone-based spraying logic.
10. Drone actuation – Control motors and spray system using PWM signals for targeted pesticide application.
11. Real-time monitoring – Update results on web interface via Firebase.
12. Stop/Repeat – Continue process for next field area or terminate system.

CHAPTER 5

RESULT

The system they made was tested in an environment that is connected to the cloud. They used a dataset called PlantVillage to see how well the system can find plant diseases and use the amount of pesticide. They used a type of deep learning model called EfficientNetB0. This model was trained for 10 rounds using a method called transfer learning. They looked at how the model worked by checking things like how accurate it was and how well it could tell the difference between different diseases. The model was really good at finding diseases. It got over 95 percent of them right. This means it can work well with types of plants and diseases.

When they looked at the results they got from training the model and from testing it they were very similar. This is because they used some techniques to stop the model from getting too good at one type of plant or disease. They also looked at a table called a confusion matrix. This table showed that the model was really good at telling the difference between diseases.. Sometimes it got a little mixed up between diseases that look similar. They used a way of looking at pictures to figure out how bad a disease was. This method is called HSV-based image segmentation. They looked at the ratio of parts of the leaf to the whole leaf.

The system can find diseases in plants well. It can even do it in lighting conditions. This is important for farming because it can help farmers use the amount of pesticide. The plant disease detection model is a part of making farming more efficient. It can find diseases faster and more accurately than people can. The system can also help farmers use pesticide. This is better for the plants and the environment. There are some things that can affect how well the system works. These things include the quality of the pictures and the weather.. If they can make the system better by using more pictures and testing it in real fields it could be really useful for farming.

The plant disease detection model can play a role in improving agricultural efficiency by identifying plant diseases faster and more accurately than traditional manual methods. The system also supports targeted pesticide application reducing chemical usage and improving crop management. Although factors such, as image quality, weather conditions and field variations may affect performance further improvements using datasets and real-time field testing can enhance system reliability and practical deployment for smart farming applications.

Table 5.1. Accuracy, Confidence and Severity Analysis

Accuracy, Confidence and Severity Analysis	Disease, Confidence, Severity, Severity Band			
	Disease Class	Confidence (%)	Severity (%)	Severity Band
Sample 1	Potato Early Bight	99.6	17.17	Medium
Sample 2	Potato Late Bight	43.60	63.55	Very High
Sample 3	Tomato Late Bight	99.03	92.04	Very High
Sample 4	Tomato Early Bight	43.60	17.71	Medium

Table 5.2. Pesticide, Dosage, Mixing, Drone Flow

Samples	Pesticide, Dosage, Mixing, Drone Flow			
	Pesticide	Dosage(ml/L)	Mixing (ml in 5L)	Drone Flow (ml/min)
Potato Early Bight	Mancozeb	2.0	10.0	1981.0
Potato Early Bight	Cymoxanil + Mancozeb	3.5	17.5	1191.0
Tomato Early Bight	Chloronthalonil	2.0	10.0	1200.0
Tomato Late Bight	Cymoxanil + Mancozeb	3.5	17.5	1200.0

```
✓ ... ✓ Firebase Re-initialized Successfully with correct region
✓ Class names loaded successfully: ['Pepper__bell__Bacterial_spot',

🔍 Checking Firebase for new image...
✓ Monitoring completed.
✓ New image downloaded: esp32_images/image_27678.jpg
1/1 _____ 0s 132ms/step
🌿 Disease: Tomato_Early_blight
📊 Severity: 93.11 %
⚠️ Severity Level: Critical
💊 Pesticide: Chlorothalonil
💧 Dosage: 4 ml per Liter
🚰 Spray ON
✓ Firebase Updated Successfully
Spray → ON, PWM → 237
Relays → ON, ON, ON, ON
```

Figure 5.1. MI Model Output

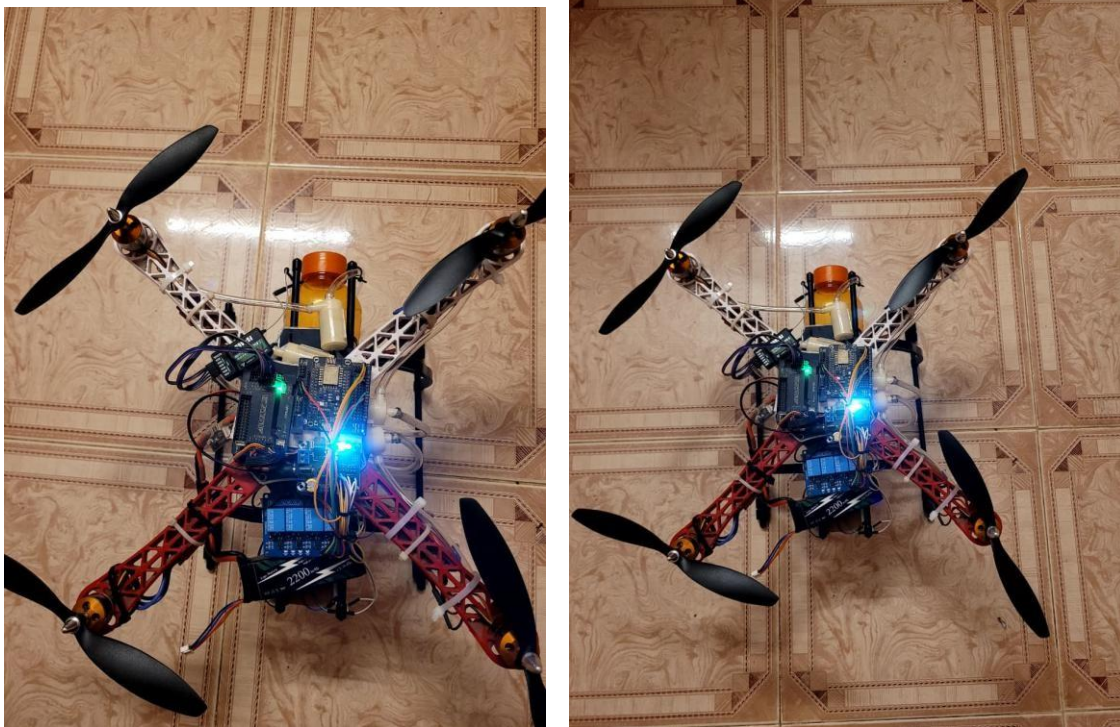


Figure 5.2. Prototype Model

CHAPTER 6

CONCLUSION

6.1 Conclusion

This research is about a drone system that helps find plant diseases and applies pesticides in the amount. The drone system uses computer programs like learning and image processing to make farming more efficient.

The drone system is really good at finding plant diseases. It uses learning and image processing to do this. The drone system is also good at applying pesticides in the amount. This makes the drone system very useful for farming.

The drone. Deep learning and image processing work together to make farming easier. The drone system is a help to farmers because it uses deep learning and image processing to find plant diseases and apply pesticides. This is what the research is, about. The drone system and how it helps with farming using learning and image processing. It has a camera that takes pictures of plants and a computer program called Convolutional Neural Network or EfficientNetB0 for short that looks at these pictures to figure out what disease the plant has.

The system is connected to the internet through a platform called Firebase, which lets all the different parts of the system talk to each other in real time. This means that the drone can take a picture of a plant the computer can look at the picture. Figure out what disease it has and the drone can apply the right pesticide, all automatically. The plant disease detection system and pesticide application system are helpful because they reduce the amount of work people must do reduce the amount of pesticide used and help plants grow better. The system is easy to use. Helps farmers take care of their plants at the right time. However, the plant disease detection system and pesticide application system do depend on data, good weather and a good internet connection to work properly.

6.2 Future Scope

The Using a wider range of datasets from different places can make the model more accurate and improve the system. Edge computing and the Internet of Things (IoT) can help the drones process data. This makes the internet less necessary. Speeds up the system's response time. Automated spraying systems are one more way to make things better. Apps that can be used in many languages and without an internet connection. These changes can make the system work

better, be more useful, and be able to grow. They can also help make farming more environmentally friendly. The updates will help both the system and the drones. Farmers will have a hard time using the drones and system. The drones and system will help farmers more. The system and drones can help more farmers. The drones and the system can help farmers even more.

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